

Linear Equations

Geometric Algorithms
Lecture 1

Outline

- » Motivating examples of linear systems
- » Formally define linear systems
- » Solve systems of linear equations

Keywords

linear equations

coefficient

unknowns

point sets

hyperplanes

systems of linear equations

consistency

augmented matrix

coefficient matrix

forward-elimination

back-substitution

row operations

row equivalence

Motivation

Lines (Slope-Intercept Form)

$$y = \underbrace{m}_\text{slope} x + \underbrace{b}_{y\text{-int.}}$$

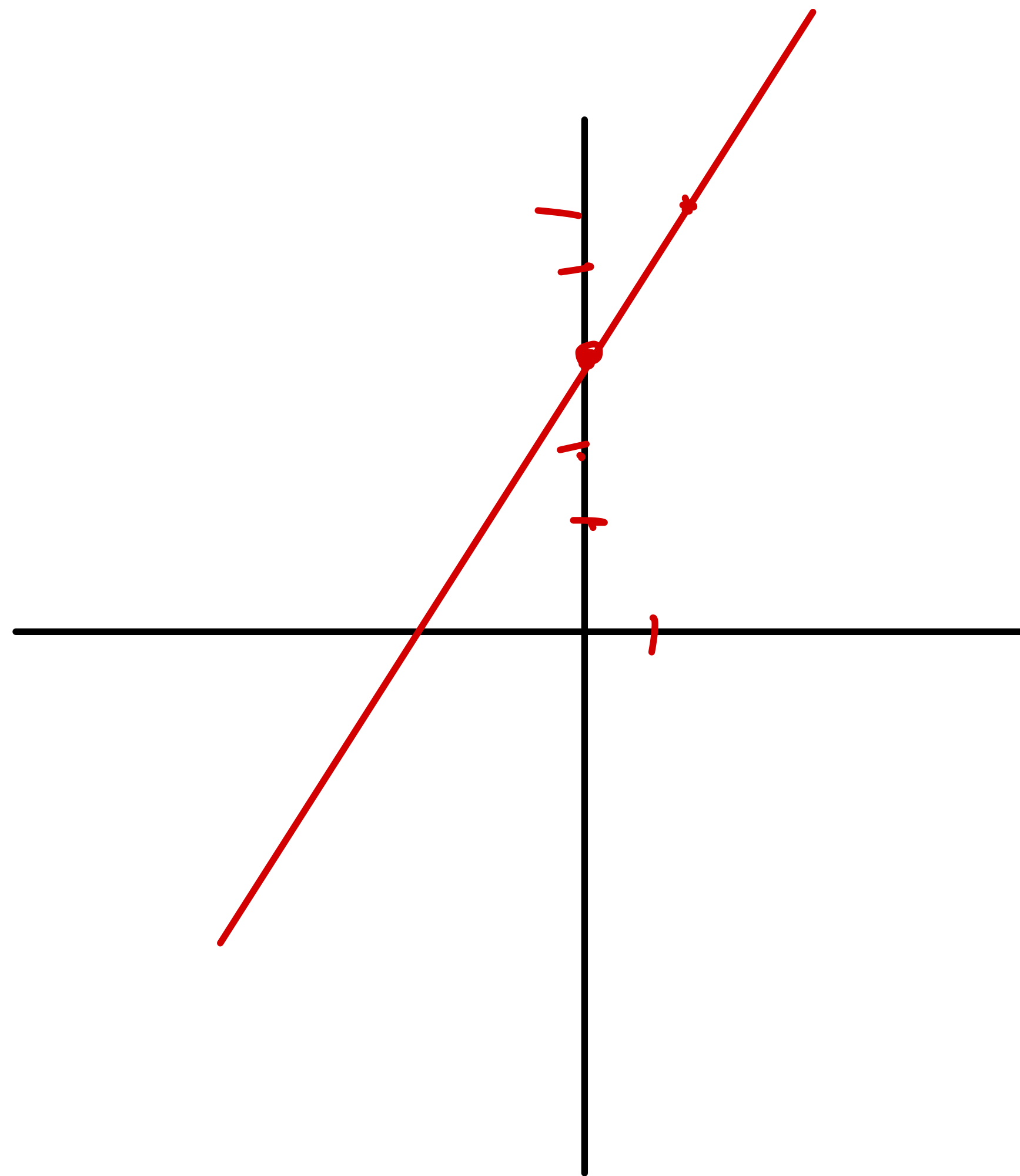
$$y = 2x + 3$$

$$x = 1$$

$$y = 2(1) + 3$$

$$(1, 5)$$

$$\begin{aligned} x &= 0 \\ y &= 3 \end{aligned}$$



Lines (Slope-Intercept Form)

$$y = mx + b$$

Lines (Slope-Intercept Form)

$$y = mx + b$$

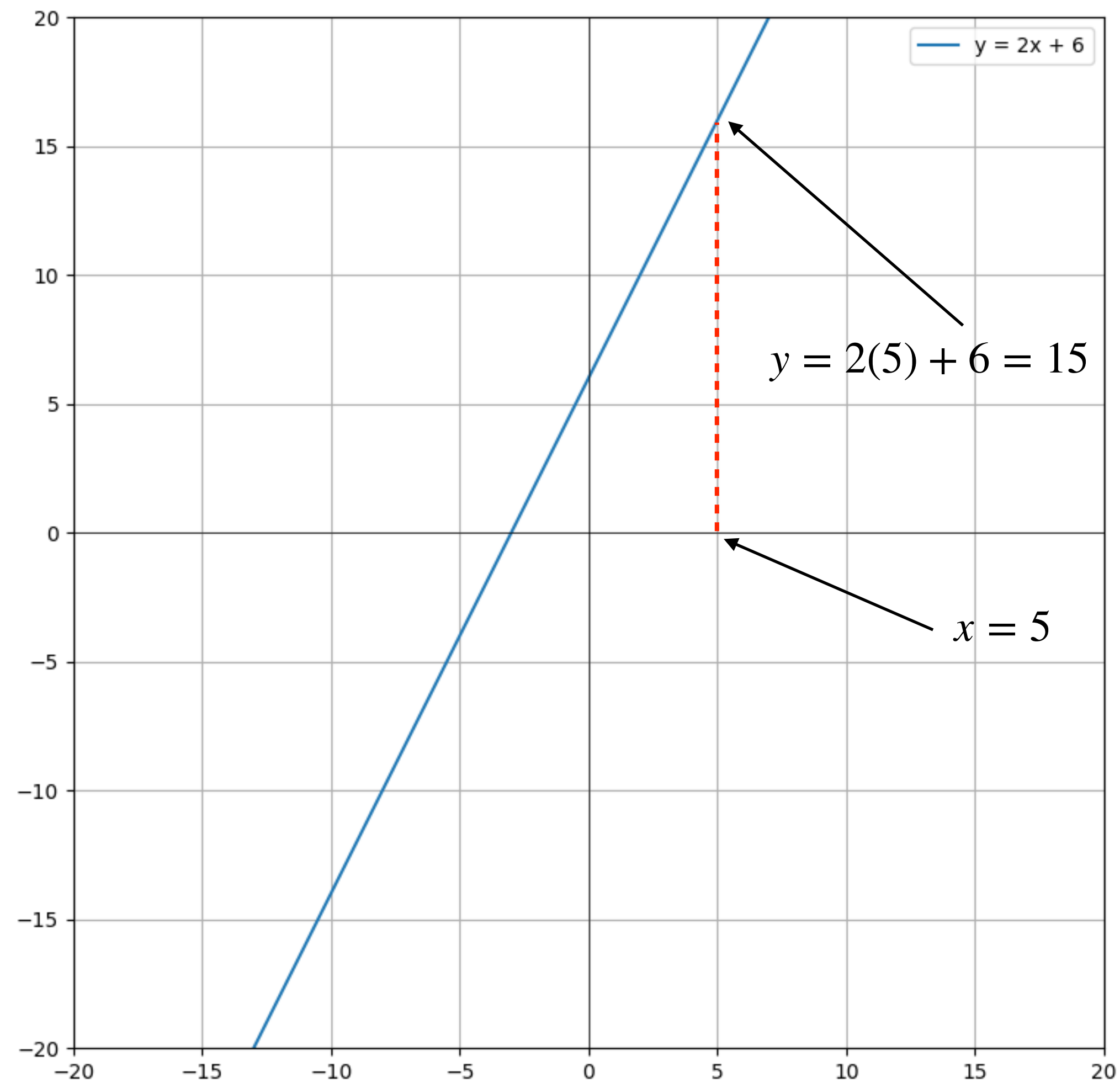
slope y-intercept

Lines (Slope-Intercept Form)

$$y = \underbrace{mx}_{\text{slope}} + \underbrace{b}_{\text{y-intercept}}$$

Given a value of x , compute a value of y

Lines (Graph)



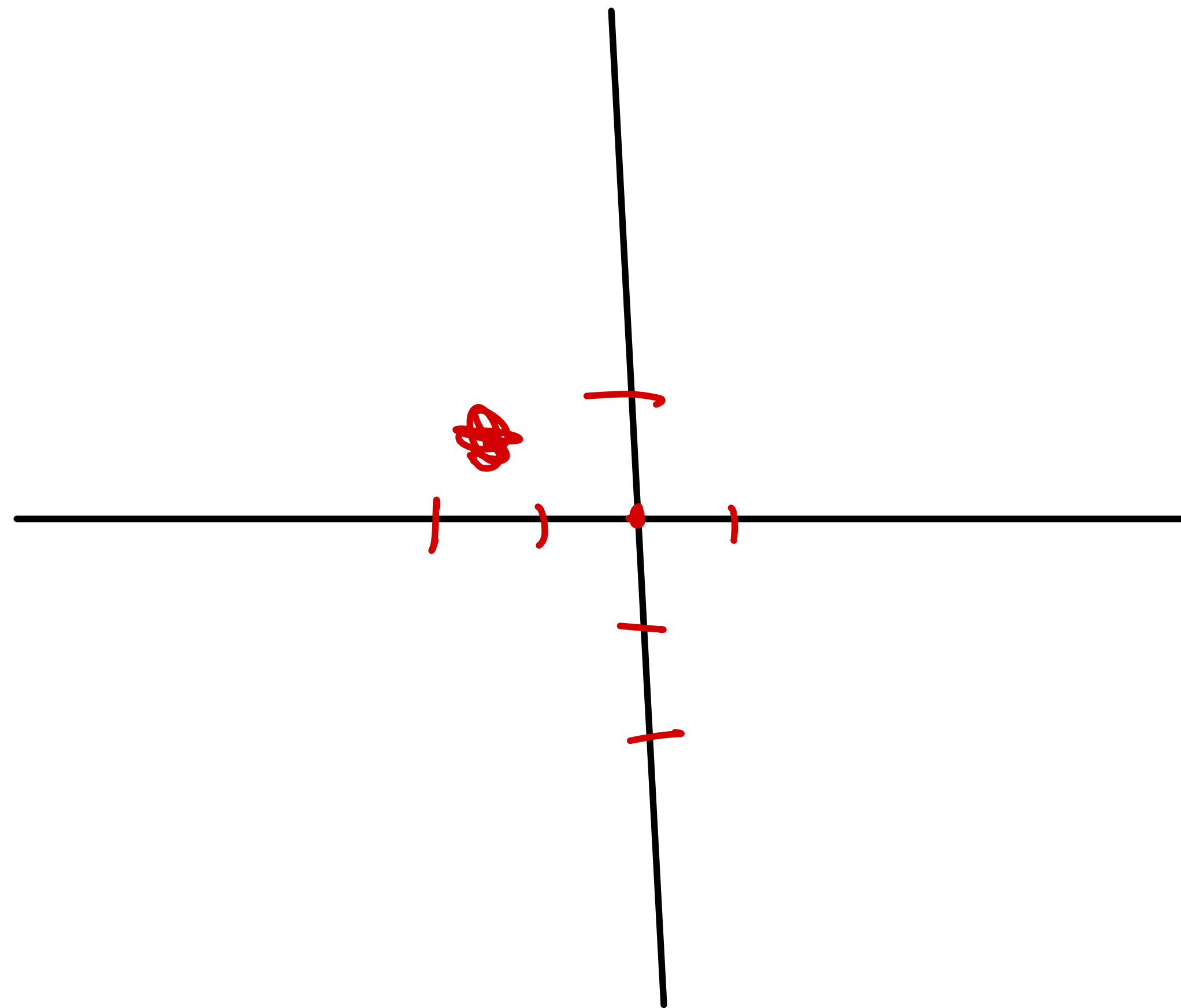
Lines (General Form)

$$c = ax + by$$

$$0 = 2x + 3y$$

$$(0, 0)$$

$$(-\frac{3}{2}, 1)$$



Lines (General Form)

$$ax + by = c$$

Lines (General Form)

$$ax + by = c$$

x-intercept: $-\frac{c}{a}$

Lines (General Form)

$$ax + by = c$$

x-intercept: $\frac{c}{a}$

y-intercept: $\frac{c}{b}$

Lines (General Form)

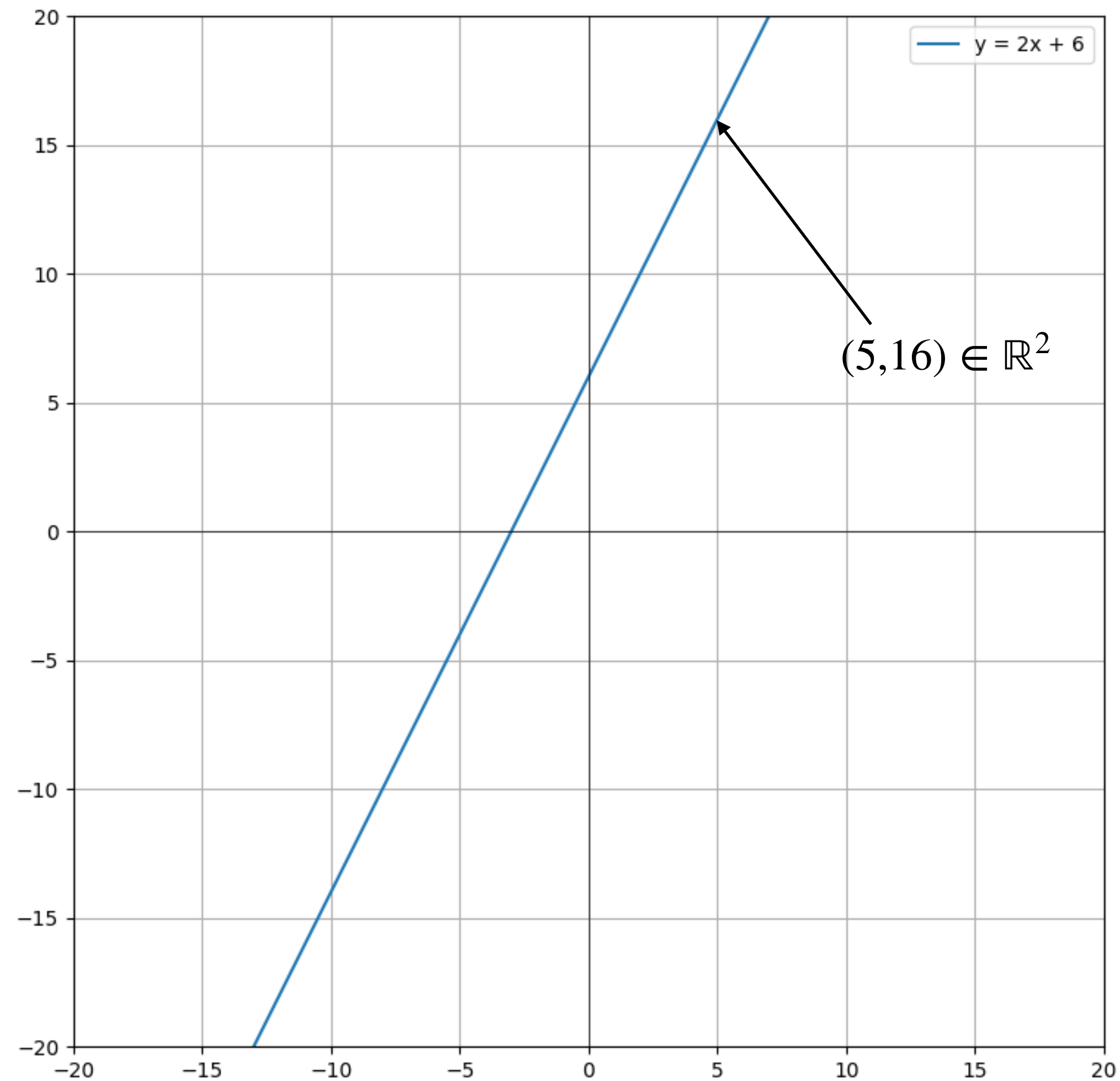
$$ax + by = c$$

$$\text{x-intercept: } \frac{c}{a}$$

$$\text{y-intercept: } \frac{c}{b}$$

What values of x and y make the equality hold?

Lines (Graph)



$$\{(x, y) : (-2)x + y = 6\}$$

Lines

slope-int $\rightarrow$ general

$$(-m)x + y = b$$

general $\rightarrow$ slope-int

$$y = \left(\frac{-a}{b} \right) x + \frac{c}{b}$$

Line Intersection

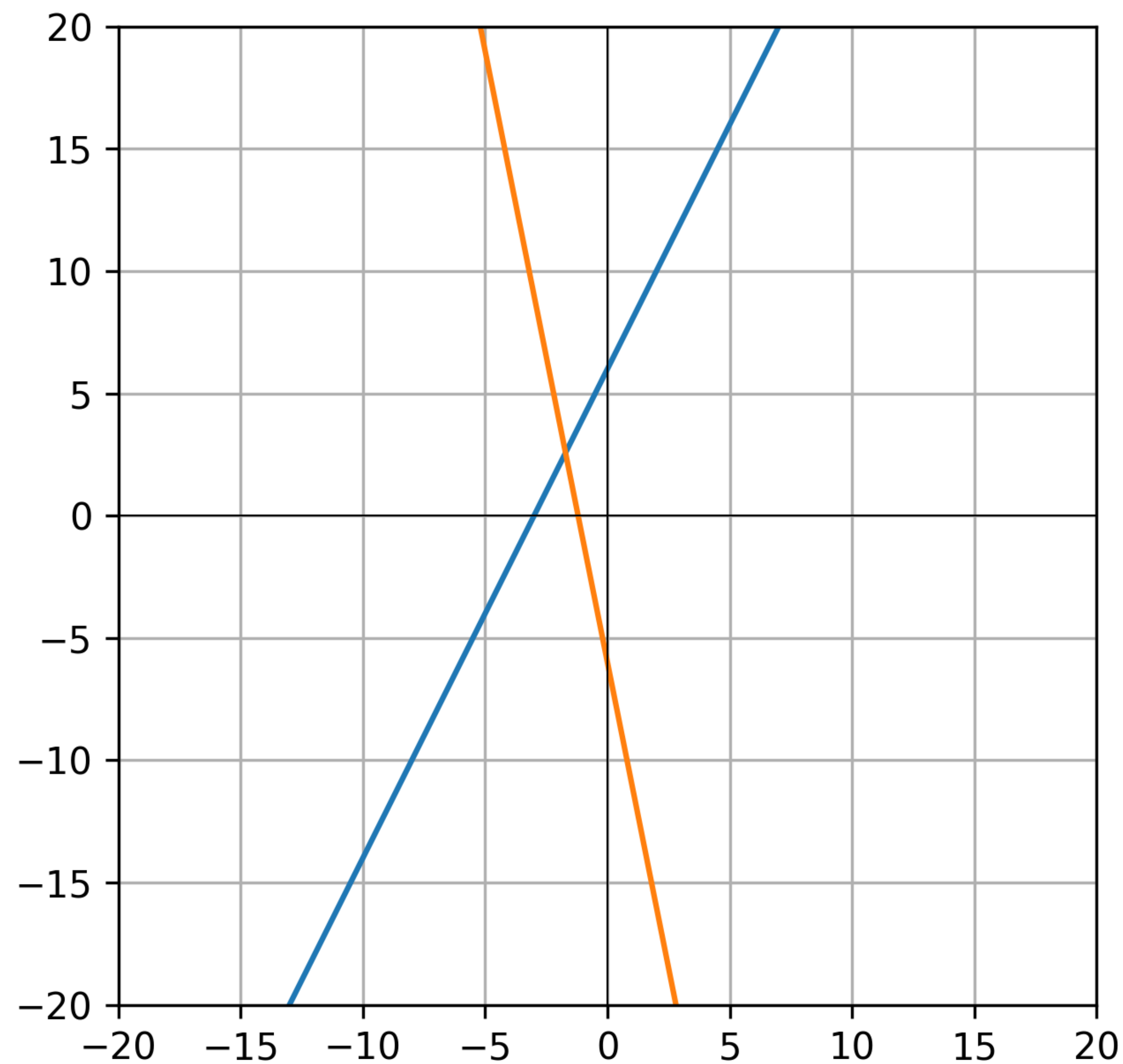
$$y = m_1x + b_1$$

$$y = m_2x + b_2$$

Q. Given two lines, where do they intersect?

Line Intersection

Line Intersection (Graph)



Line Intersection (Alternative)

Line Intersection (Alternative)

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Q. Given two (general form) lines, what values of x and y satisfy ***both*** equations?

Line Intersection (Alternative)

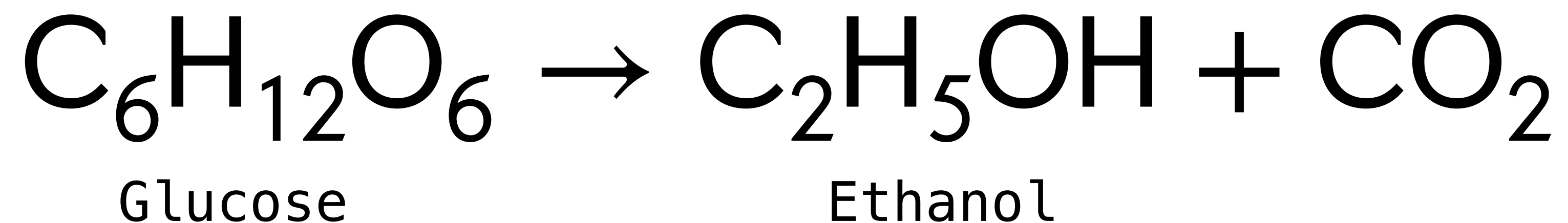
$$a_1x + b_1y = c_1$$

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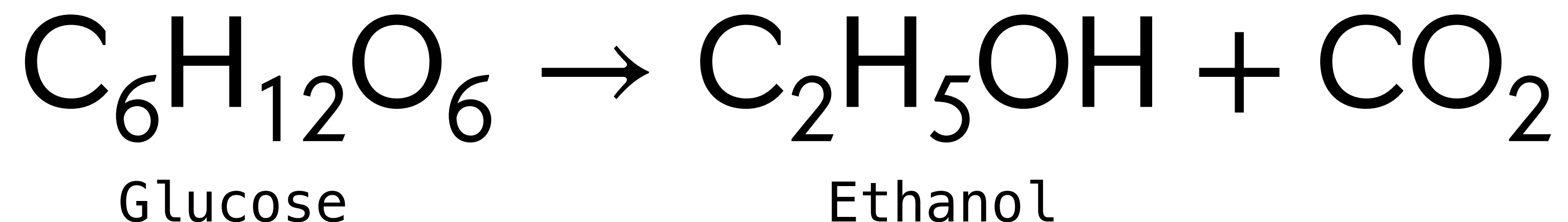
Q. Given two (general form) lines, what values of x and y satisfy ***both*** equations?

This is the same question

Example: Balancing Chemical Equations

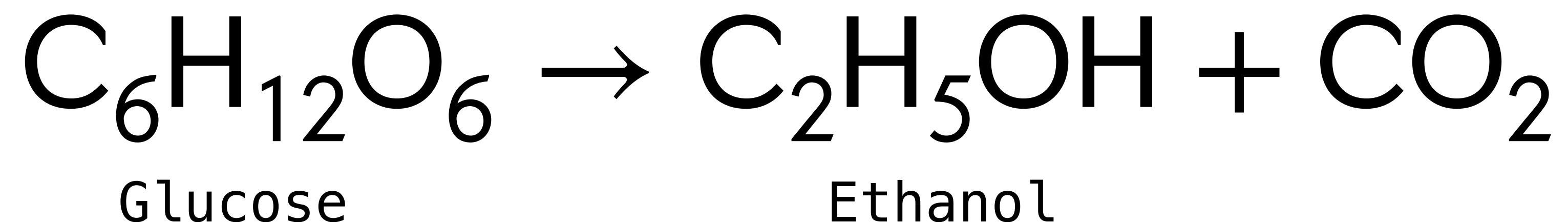


Example: Balancing Chemical Equations



How much ethanol is produced by fermentation?
(for science)

Example: Balancing Chemical Equations



How much ethanol is produced by fermentation?
(for science)

The **number of atoms** has to be *preserved* on each side of the equation

Balancing Chemical Equations



Balancing Chemical Equations



$$6\alpha = 2\beta + \gamma \quad (\text{C})$$

$$12\alpha = 6\beta \quad (\text{H})$$

$$6\alpha = \beta + 2\gamma \quad (\text{O})$$

Balancing Chemical Equations



$$6\alpha - 2\beta - \gamma = 0 \quad (\text{C})$$

$$12\alpha - 6\beta = 0 \quad (\text{H})$$

$$6\alpha - \beta - 2\gamma = 0 \quad (\text{O})$$

Formal Definitions

Linear Equations

$$2x + 3y + 4z + 5w = 6 \quad \checkmark$$

$$2x^2 = 3 \quad \times$$

$$xy + z = 4 \quad \times$$

$$2\bigcirc x + 3y = \bigcirc \quad \times$$

Linear Equations

Df. A *linear equation* in variables $x_1, x_2, \dots, x_n$ is an equation which can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n, b$ are real numbers ($\mathbb{R}$)

Linear Equations

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coefficients

$$\boxed{a_1}x_1 + \boxed{a_2}x_2 + \dots + \boxed{a_n}x_n = b$$

where $a_1, a_2, \dots, a_n, b$ are real numbers ($\mathbb{R}$)

Linear Equations

$$3x = 2y \quad \checkmark$$

Df. A *linear equation* in variables $x_1, x_2, \dots, x_n$ is an equation which can be written in the form

$$\overset{\text{coefficients}}{a_1} \overset{\text{unknowns}}{x_1} + a_2 x_2 + \dots + a_n x_n = b$$

where $a_1, a_2, \dots, a_n, b$ are real numbers ($\mathbb{R}$)

Point sets

Point sets

Linear equations describe *point sets*:

$$\{(s_1, s_2, \dots, s_n) \in \mathbb{R}^n : a_1 s_1 + a_2 s_2 + \dots + a_n s_n = b\}$$

Point sets

Linear equations describe *point sets*:

Euclidean Space

$$\{(s_1, s_2, \dots, s_n) \in \mathbb{R}^n : a_1s_1 + a_2s_2 + \dots + a_ns_n = b\}$$

The points such that the equation holds

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

Not a line...

Demo

$$z = 5$$

$$0x + 0y + z = 5$$

What does the point set of this linear equation look like?

$\{ (x, y, z) \in \mathbb{R}^3 \text{ s.t. } z = 5 \}$

space

$$(3, 4, 5) \quad (0, 0, 5)$$

$$(1, 2, 5)$$

Demo

$$-x + 0y + z = 5$$

How about this one?

$$\{ (x, y, z) \in \mathbb{R}^3 \text{ s.t. } z = 5 + x^2 \}$$

$$(2, 0, 7)$$

(Note: The value 2 is circled in red in the original image)

$$(6, 7, 11)$$

(Note: The value 6 is circled in red in the original image)

Demo

$$-x + -y + z = 5$$

How about this one?

Hyperplanes

Hyperplanes

The point set of a general linear equation
is called a **hyperplane**

Hyperplanes

The point set of a general linear equation is called a **hyperplane**

Theme of the course: Hyperplanes "behave" like lines and planes in many respects

Linear Equations, Geometrically

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Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

A plane!

Systems of Linear Equations

Df. A *system of linear equations* is just a collection of linear equations *over the same variables*

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Df. A *system of linear equations* is just a collection of linear equations *over the same variables*

Df. A *solution* to a system is a point that satisfies all its equations *simultaneously*

Example

$$x + 2y = 1$$

$$-x - y - z = -1$$

$$2x + 6y - z = 1$$

Show that $(3, -1, -1)$ is solution to the above linear system

$$3 + (2)(-1) = 1 \quad \checkmark$$

$$-3 - (-1) - (-1) = -1 \quad \checkmark$$

$$2(+3) + 6(-1) - (-1) = 1 \quad \checkmark$$

System of Linear equations (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

System of Linear equations (General-form)

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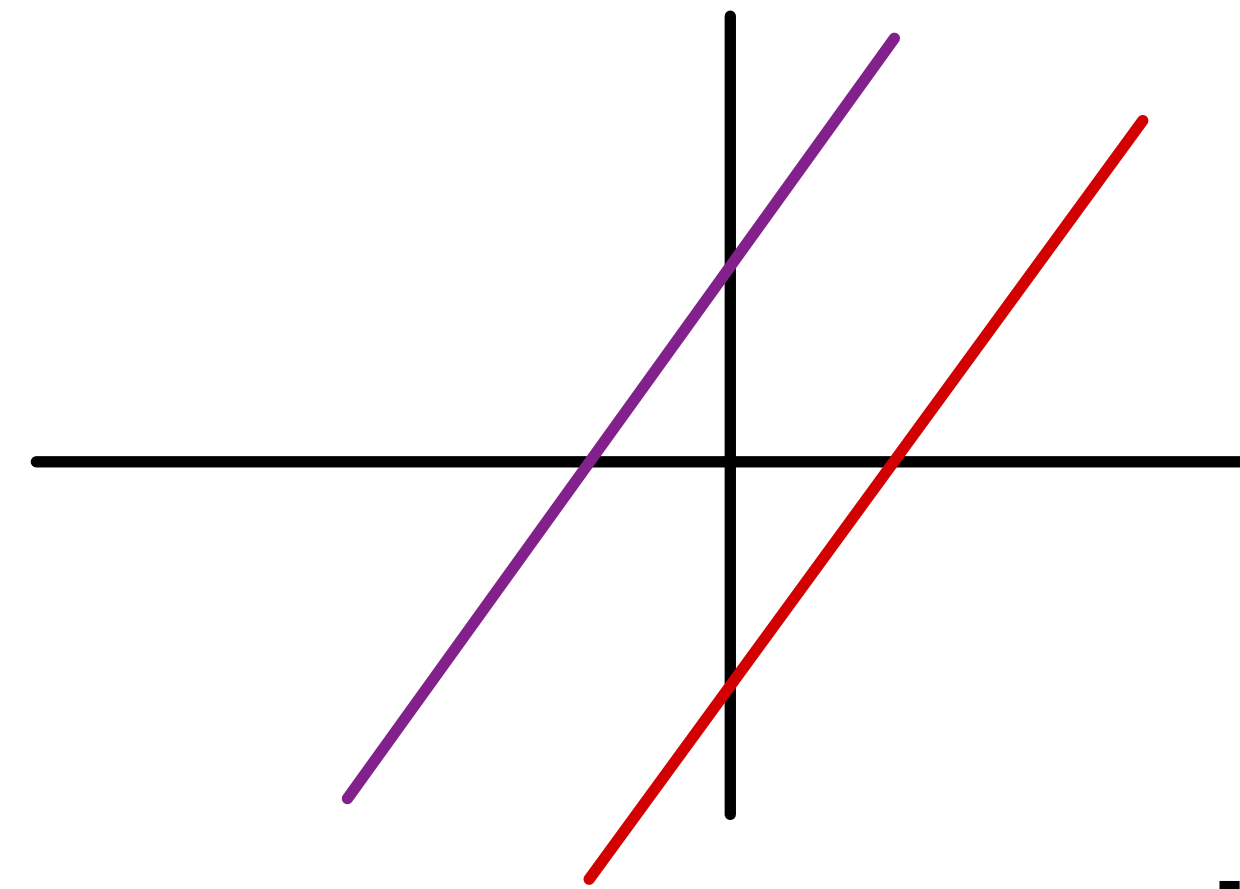
Does a system have a solution?

How many solutions are there?

What are its solutions?

Consistency

Consistency



Df. A system of linear equations is ***consistent*** if it has a solution

It is ***inconsistent*** if it has no solutions

demo

(consistency/inconsistency in $\mathbb{R}^3$)

Number of Solutions

Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

These are the **only** options

Matrix Representations

$$2x + 3y = -8$$

$$y = 2$$

$$2y = 0$$

Matrix Representations

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \approx \begin{array}{l} 2x + 3y = -8 \\ y = 2 \\ 2y = 0 \end{array}$$

Matrix Representations

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Writing down the unknowns is *tedious* (and more difficult to input into a computer)

We'll write down linear systems as **matrices**, i.e., 2D grids of numbers with *fixed* width and height

a matrix is just a representation

Matrix Representations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Matrix Representations

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augmented matrix

Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

coefficient matrix

Matrix Representations

$$6\alpha - 2\beta - \gamma = 0 \quad (\text{C})$$

$$12\alpha - 6\beta = 0 \quad (\text{H})$$

$$6\alpha - \beta - 2\gamma = 0 \quad (\text{O})$$

Matrix Representations

$$\begin{bmatrix} 6 & -2 & -1 & 0 \\ 12 & -6 & 0 & 0 \\ 6 & -1 & -2 & 0 \end{bmatrix}$$

Solving Linear Systems

Solving Systems with Two Variables

$$2x = -6 - 3y$$

$$x = -3 - \frac{3}{2}y$$

$$4(-3 - \frac{3}{2}y) - 5y = 10$$

$$-12 - 6y - 5y = 10$$

$$-11y = 22$$

$$y = -2$$

$$2x + 3y = -6$$

$$4x - 5y = 10$$

$$2x + 3(-2) = -6$$

$$2x - 6 = -6$$

$$x = 0$$

$$4(0) - 5(-2) = 10 \quad \checkmark$$

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$2x = (-3)y - 6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$4((-3/2)y - 3) - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$-6y - 12 - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$-11y = 22$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)(-2) - 3$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = 3 - 3$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = 0$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Question

This works, but *why might we want a different approach?*

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Another Approach

Elimination

Eliminate x from the EQ2 and solve for y

Eliminate y from EQ1 and solve for x

Back-Substitution

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$



$$2x + 3y = 6$$

$$-11y = 22$$



$$2x + 3y = 6$$

$$y = -2$$

forward elimination.

$$2x + 3y = -6$$

$$4x - 5y = 10$$

$$\begin{array}{r} -2(2x) \quad -2(3y) \quad -2(-6) \\ \hline \end{array}$$

$$\begin{array}{r} \textcircled{0} \rightarrow -11y = 10 + 12 \\ -11y = 22 \end{array}$$

$$y = -2$$

Motivating Question:
How does this look with matrices?

Solving Systems as Matrices

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the *same solutions*

Solving Systems as Matrices

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the *same solutions*

Can we represent these intermediate steps as operations on matrices?

Let's look back at this...

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Elementary Row Operations

scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

Elementary Row Operations

scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

These operations don't change the solutions

Scaling Example

Scaling Example

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_1 \leftarrow 2R_1$$


$$\begin{array}{l} 4x + 6y = -12 \\ 4x - 5y = 10 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 6 & -12 \\ 4 & -5 & 10 \end{bmatrix}$$

Replacement Example

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_2 \leftarrow R_2 + R_1$$



$$\begin{array}{l} 2x + 3y = -6 \\ 6x - 2y = 4 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 6 & -2 & 4 \end{bmatrix}$$

Replacement Example

Interchange Example

$$\begin{aligned} 2x + 3y &= -6 \\ 4x - 5y &= 10 \end{aligned}$$

$$R_1 \leftrightarrow R_2$$



$$\begin{aligned} 4x - 5y &= 10 \\ 2x + 3y &= -6 \end{aligned}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -5 & 10 \\ 2 & 3 & -6 \end{bmatrix}$$

Interchange Example

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

Example: Row Reductions

$$\begin{array}{ccc} & R_2 \leftarrow R_2 - 2R_1 & \\ \begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} & \longrightarrow & \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix} \\ & R_2 \leftarrow R_2 / (-11) & \\ & \longrightarrow & \begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix} \end{array}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / (-11)$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



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$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 / 2$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

elimination

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

back-substitution

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Row Equivalence

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

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$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

We can compute solutions by sequence of row operations

Question

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How do we know when we're done? What is the "target" matrix?

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When the final matrix "looks like" a solution

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How do we know when we're done? What is the "target" matrix?

When the final matrix "looks like" a solution

We'll get more into that next time...

Example

$$x - 2y + z = 5$$

$$2y - 8z = -4$$

$$6x + 5y + 9z = -4$$

Summary

Linear equations define *hyperplanes*

Systems of linear equations may or may not have *solutions*

Linear systems can be represented as *matrices*, which makes them more convenient to solve